

# Reshaping Access to the Past

Computer Vision and RAG in Historical Archives

LA7  
FONERIA

A FUTURE  
FOR THE  
PAST

Gerard Corbella



# How long to describe what we have already digitised?

1,562  
*years*

estimated time to fully describe the 40 million photographic documents held in Catalan archives — at current, manual archival rates.



**And that is only the photographs.**

Add millions of pages of military court records, local censuses and countless other series.

A fragment of history — victims, democratic memory — condemned to a millennium of invisibility, purely by the human cost of transcription.

***The bottleneck is not storage. It is description.***

# AI learns from a thin, recent slice of the web — while humanity's record stays invisible

## WHAT LLMs ARE TRAINED ON

**60–80 M**

machine-readable web pages

### **Recent. Partial. Unverified.**

Riddled with bias and misinformation, and exposed to manipulation of search and ranking.

## WHAT ARCHIVES PRESERVE

**5,500 yrs**

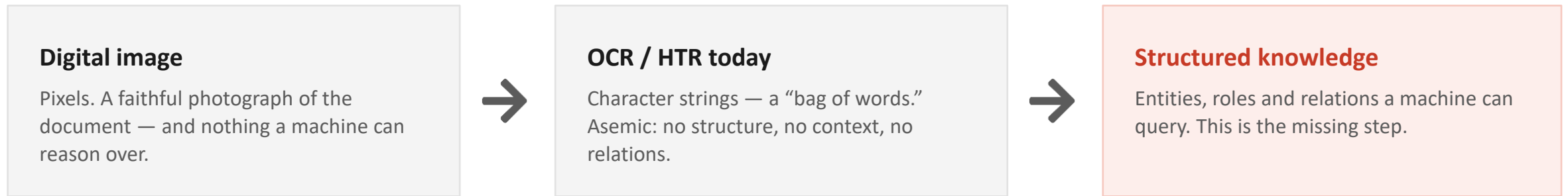
of recorded human knowledge

### **Primary sources. Trustworthy.**

Europe alone holds well over a billion records — yet only ~8% is online, and to machines it is still just pixels.

# From the digital image to structured knowledge

Twenty years of digitisation prioritised capture and storage. But a scanned parchment is not knowledge — it is a picture of knowledge.



**Layout is semantic, not linear.** Newspaper columns, marginal annotations, signatures validating a paragraph, handwritten amendments over typescript — sequential reading turns them into an unreadable “soup of letters.” A testament only means “testament” if the system grasps the roles (testator, heir, notary), wherever the names appear.

# From extracting text to understanding the document

## 01 Datafication

*making the document machine-readable*



**HTR & OCR** handwriting, print, typescript



**Layout analysis** columns, tables, reading order



**NER** names, places, organisations



**Photo dating** inferring date & context from visual cues

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Computer Vision · the CVC's “deep tech”



## 02 Semantization

*giving the data formal & relational meaning*



**Records in Contexts (RiC-O)** a native semantic data model for archives



**Entity linking** relations across documents



**Knowledge graph** nodes & edges, not flat records



**Archival validity** roles, provenance, context preserved

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Archival science · La Foneria's humanistic layer

# Why Records in Contexts (RiC-O)



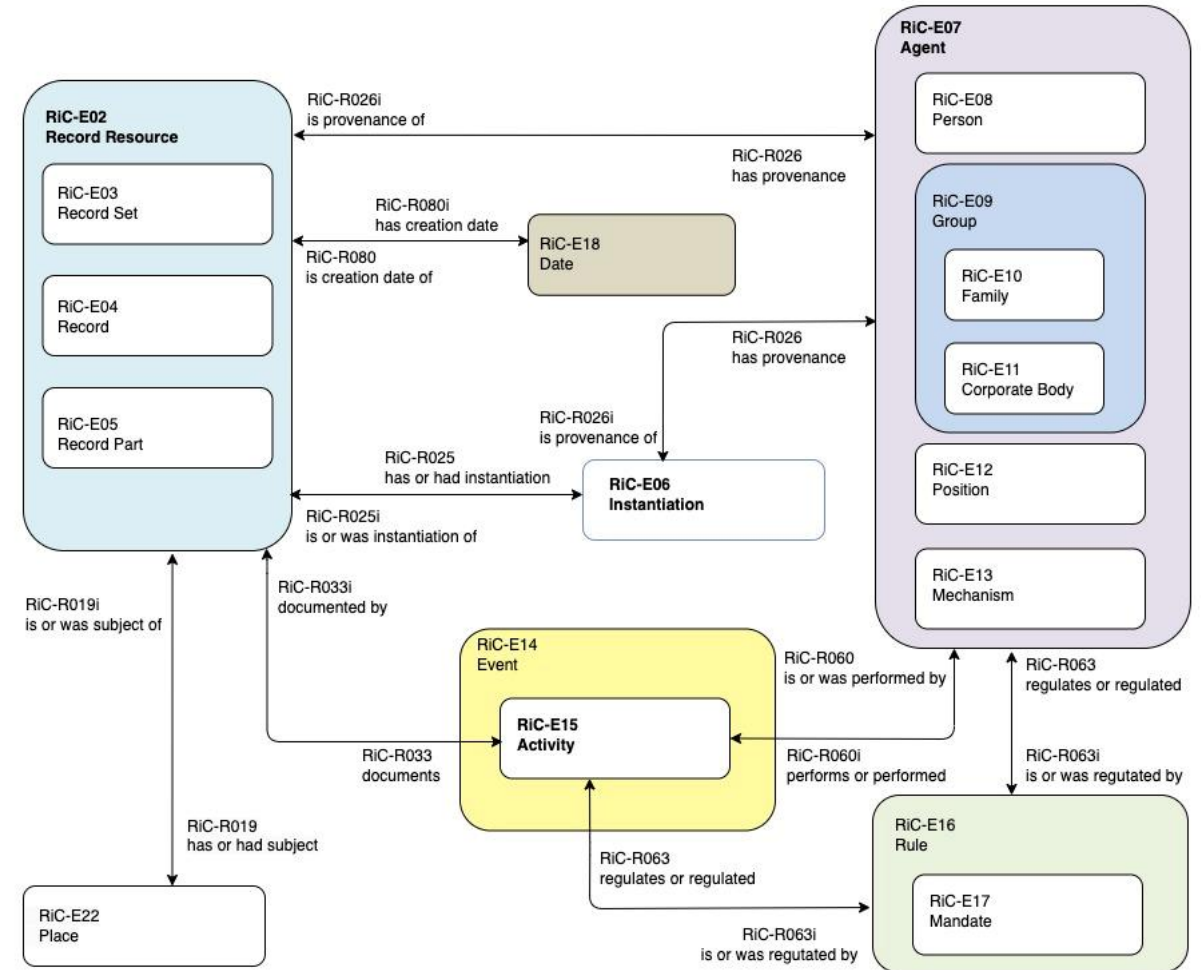
The ICA / EGAD ontology that adds the semantic layer ISAD(G) lacks.

Instead of flat, isolated descriptions, RiC-O models records, agents, activities, places and rules as a connected graph.

Adopted as the native data model — not bolted on afterwards — so AI output is archivally valid and interoperable by design (RDF, FAIR, the 5-star open-data path).

**We cannot build a new database for every archival series.** We need one model with semantic capacity. RiC-O is that bridge between AI and archival science.

RiC-CM v1.0: a global overview

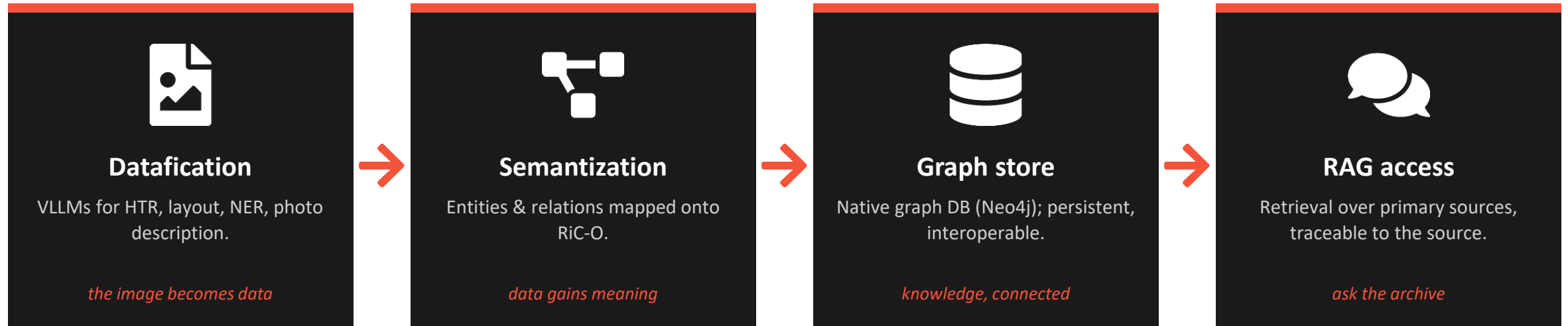


Created by ICA-EGAD in September 2023.

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# The architecture: from pixels to a question you can ask



**RAG is the antidote to hallucination.** Unlike commercial LLMs working over the open web, retrieval runs directly on primary archival sources — every answer is traceable to the original document, preserving chain of custody and **trustworthiness**. Open models (e.g. AINA) keep it working in Catalan.

# Research challenges when you take this to real archives



## Unified multimodality

Manuscripts, press and photographs unified in a single semantic representation — so they can be related, not siloed per tool.



## Relations at scale

No prior work on discovering and managing semantic relations across very large, heterogeneous document collections.



## NL ↔ knowledge graphs

Translating natural-language questions into queries over knowledge graphs, in relevant, massive settings.



## Logical-spatial understanding

Reconstructing the “news” unit in multi-column press; linking signatures, paragraphs and marginal notes in manuscripts.



# Democratic memory & transitional justice

*Where the archive is the only reliable source to locate the dead and the disappeared — and where there are never enough hands to read it.*



A semantic-capable archive answers questions, not keywords:



Who was this victim — and what was their family network?



How did the sentence evolve over time?



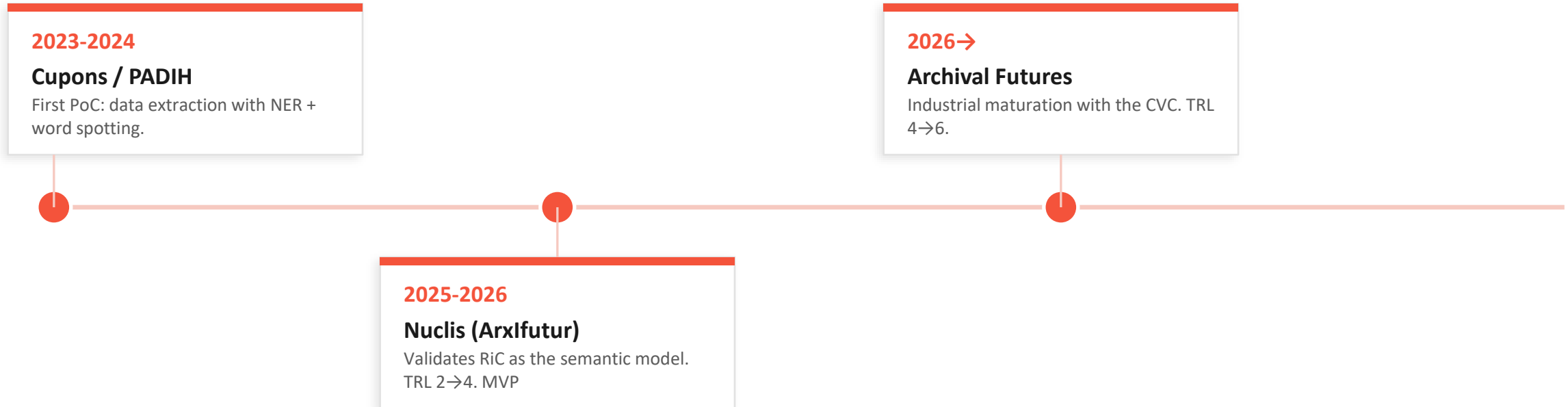
Which victims were held in the same prison?



Which members of a political party were repressed?

**NER alone cannot do this. It finds names, but it cannot tell if is a victim or a perpetrator. Semantic understanding can.**

# This is grounded in real research, not speculation



*A standing partnership with the CVC — Document Analysis Group (DocAI) — across the whole TRL ladder.*

# Archives stop being repositories — they become data providers

Users no longer hunt for documents. They ask questions. The archive becomes a trustworthy substrate for the semantic web and for training future, domain-specific AI.



## Natural-language access

Question–answer interaction with primary sources, traceable to the original.



## New research power

Who, when, what, how — at a scale manual description could never reach.



## Reusable services

Anonymisation, automatic description, dating — as interoperable web services.



## Digital storytelling

Online exhibitions and educational products built on structured memory.

# A future for the past

Computer vision reads the past. Semantics gives it meaning. RAG lets us ask it questions.

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La Foneria · in collaboration with the Computer Vision Center (CVC)

*Thank you — questions welcome*